

In-Course Assessment Examination – 03
IT1214(P) – Object Oriented Programming
Department of Physical Science
Answer all Questions

Allowed Time: 60 Minutes

June 2024

Instructions:

- Read the questions carefully and provide the appropriate Java code to solve the problem.

Question 1:

You are asked to design a Cricket Application, where you need to implement the classes to represent batsmen and bowlers. Follow the instructions below:

a) Create an abstract class named "**Player**" with the following attributes:

- **playerID** (int): to store the unique ID of the player
- **name** (String): to store the name of the player
- **country** (String): to store the country of the player
- **scores** (int []): a 1D array to store the scores made by the player

Include the following methods in the "**Player**" class:

- **Constructor**: to initialize the playerID, name, and scores.
- **getPlayerID()**: a method to retrieve the player's ID.
- **getName()**: a method to retrieve the player's name.
- **getCountry()**: a method to retrieve the player's country.
- abstract void **addScore**(int score): an abstract method to add a score to the player.
- abstract void **printDetails()**: an abstract method to print the details of the player.

b) Create a subclass named "**Batsman**" that inherits from the "**Player**" class.

The "**Batsman**" class should have an additional attribute:

- **runs** (int): to store the total runs made by the batsman

Implement the following methods in the "**Batsman**" class:

- **Constructor**: to initialize the playerID, name, and scores. Calculate the total runs from the scores.
- **getRuns()**: a method to retrieve the total runs made by the batsman.
- Override the **addScore(int score)** method to add the new score to the scores array.
Update the total runs.
Condition: A score value must be between 0 and 100
- Override the **printDetails()** method to print the batsman's ID, name, country, scores, and total runs.

c) Create a subclass named "Bowler" that inherits from the "Player" class.

The "Bowler" class should have an additional attribute:

- **wickets** (int): to store the total wickets taken by the bowler

Implement the following methods in the "Bowler" class:

- **Constructor**: to initialize the playerID, name, and scores. Calculate the total wickets from the scores.
- **getWickets()**: a method to retrieve the total wickets taken by the bowler.
- Override the **addScore(int score)** method to add the score to the scores array
Update the total wickets.
Condition: Score must be between 0 and 10
- Override the **printDetails()** method to print the bowler's ID, name, country, scores, and total wickets.

Question 2:

Create an array of 5 objects for the "**Batsman**" class and an array of 5 objects for the "**Bowler**" class. Follow the instructions below:

a) Initialize the objects with the following details:

- Batsman Objects:
 - Batsman 1: ID = 1, Name = "Sachin Tendulkar", "India", Scores = {42, 0, 61, 15, 0, 100, 23, 64, 41, 0, 26}
 - Batsman 2: ID = 2, Name = "Virat Kohli", "India", Scores = {0, 25, 10, 60, 11, 0, 40, 11, 0, 2, 0}
 - Batsman 3: ID = 3, Name = "MS Dhoni", "India", Scores = {6, 22, 48, 0, 11, 69, 0, 21, 19, 0, 42}
 - Batsman 4: ID = 4, Name = "Ricky Ponting", "Australia", Scores = {11, 0, 28, 0, 64, 18, 45, 26, 0, 11, 0}
 - Batsman 5: ID = 5, Name = "Brian Lara", "West Indies", Scores = {0, 13, 21, 47, 0, 69, 17, 0, 12, 0, 40}
- Bowler Objects:

- Bowler 1: ID = 1, Name = "Muttiah Muralitharan", "Sri Lanka", Scores = {0, 10, 6, 8, 0, 1, 0, 2, 4, 0, 6}
- Bowler 2: ID = 2, Name = "Shane Warne", "Australia", Scores = {6, 0, 1, 0, 2, 0, 4, 10, 0, 6, 8}
- Bowler 3: ID = 3, Name = "Wasindu Hasaranga", "Sri Lanka", Scores = {0, 1, 0, 6, 8, 4, 2, 0, 6, 0, 10}
- Bowler 4: ID = 4, Name = "Glenn McGrath", "Australia", Scores = {4, 2, 6, 0, 10, 0, 1, 8, 6, 0, 0}
- Bowler 5: ID = 5, Name = "Dale Steyn", "South Africa", Scores = {0, 4, 0, 2, 1, 6, 10, 0, 8, 6, 0}

b) Print the details of all the batsmen and bowlers using the "**printDetails()**" method.

Question 3:

1. Add Score 14 to Batsman 1.
2. Print the details of the Batsman with the highest Runs.
3. Print the details of the Bowler with the highest Wickets.
4. Print the details of the Players from Australia.

[100 Marks]